

PHY 115L

Finite Square-Well

Michael Mulhearn

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1 Introduction and System of Units

In the previous homework, we numerically integrated the Time-Independent Schrödinger Equation (TISE):

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(x) \psi(x) = E \psi(x) \quad (1)$$

using the Fourth-Order Runge-Kutta technique. We adopted a system of units based on a characteristic length scale a and energy E_0 with

$$E_0 \equiv \frac{\hbar^2}{2ma^2} \quad (2)$$

In our computer programs, we set $a = 1$ and $E_0 = 1$, which is equivalent to reporting lengths in units of a and energy in units of E_0 .

In this assignment, we'll apply the same techniques to determine the wave function for bound states of the finite square well.

2 Bound States of the Finite Square Well

Consider the finite square well potential:

$$\frac{V(x)}{E_0} = \begin{cases} -z_0^2, & -a \leq x \leq a \\ 0 & \text{otherwise} \end{cases}$$

Where z_0^2 is a dimensionless constant, nicely consistent with our system of units, that sets the depth of the well (in units of E_0). As the potential is finite everywhere, we expect both ψ and $d\psi/dx$ to be continuous. The bound state potentials will have

$$-z_0^2 < \frac{E}{E_0} < 0.$$

For $x < -a$ and $x > a$, the TISE becomes (using our system of units):

$$a^2 \frac{d^2\psi}{dx^2} = -\frac{E}{E_0} \psi(x) = (\kappa a)^2 \psi(x), \quad \kappa \equiv \frac{1}{a} \sqrt{\frac{-E}{E_0}}.$$

with the solutions

$$\psi(x) = Ae^{\kappa x}$$

for $x < -a$ and

$$\psi(x) = B e^{-\kappa x}$$

for $x > a$. Within the potential well at $-a \leq x \leq a$ we have:

$$a^2 \frac{d^2 \psi}{dx^2} = - \left(z_0^2 + \frac{E}{E_0} \right) \psi(x) = -k^2 \psi(x), \quad k \equiv \frac{1}{a} \sqrt{z_0^2 + \frac{E}{E_0}}.$$

with general solution:

$$\psi(x) = C \sin(kx) + D \cos(kx)$$

At this point, we will save ourselves a lot of hassle by noting that $V(x)$ is even, and so the general solutions can be constructed as either even or odd solutions¹ So we need only establish continuity for $x \geq 0$ and we will automatically have it for $x \leq 0$.

Starting with the even solutions and $x \geq 0$, we write:

$$\psi(x) = \begin{cases} B e^{-\kappa x} & x > a \\ D \cos(kx) & 0 \leq x \leq a \end{cases}$$

and continuity implies:

$$B e^{-\kappa a} = D \cos(ka) \quad (3)$$

We see that the dimensionless quantities κa and ka are of interest, so it will simplify things to note that:

$$(ka)^2 + (\kappa a)^2 = z_0^2$$

The derivative is:

$$\frac{d\psi}{dx} = \begin{cases} -\kappa B e^{-\kappa x} & x > a \\ -k D \sin(kx) & 0 \leq x \leq a \end{cases}$$

which both must be continuous at $x = a$, so:

$$\begin{aligned} -\kappa B e^{-\kappa a} &= -k D \sin(ka) \\ \kappa B e^{-\kappa a} &= k D \sin(ka) \\ (\kappa a) B e^{-\kappa a} &= (ka) D \sin(ka) \end{aligned}$$

where we have multiplied by a so that the equation is in terms of the dimensionless constants ka and κa . Dividing by Equation 3 we obtain:

$$\begin{aligned} \kappa a &= (ka) \tan(ka) \\ \sqrt{z_0^2 - (ka)^2} &= (ka) \tan(ka) \\ \sqrt{z_0^2 - z^2} &= z \tan(z) \end{aligned}$$

where we have defined

$$z \equiv ka = \sqrt{z_0^2 + \frac{E}{E_0}}$$

¹See Griffith's P2.1c. Or note that if $\psi(x)$ is a solution, so is $\psi(-x)$, and we can add or subtract the two to get even and odd solutions.

and so finally:

$$\sqrt{\left(\frac{z_0}{z}\right)^2 - 1} = \tan(z) \quad (4)$$

This is an equation with no known algebraic solution. It can be solved graphically or numerically. See Fig. 1. There are some interesting limiting cases to consider. When:

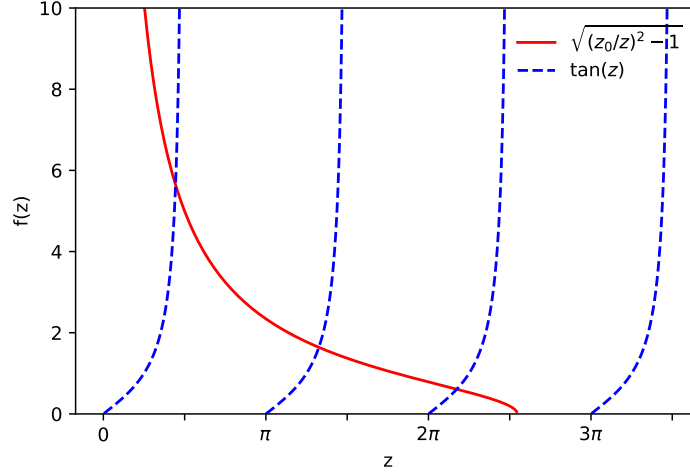


Figure 1: The allowed energies for the finite potential occur at the intersection of these curves.

$$z_0 < \frac{\pi}{2}$$

there is only one bound state (we haven't found the odd solutions yet, which is why this is $\pi/2$ and not π , as you might expect from the figure...) Recalling the definition for z_0 , this amounts to:

$$V_0 a^2 < \frac{\pi^2 \hbar^2}{8m}$$

which defines a shallow and narrow well.

Alternative, if z_0 is very large, then the intersection points are at:

$$z = \frac{n\pi}{2}, \quad n = 1, 3, 5, \dots$$

which means the allowed energies are:

$$E + V_0 = \frac{n^2 \pi^2 \hbar^2}{2m(2a)^2}$$

which (relative to the bottom of the well) are the energies of the infinite square well.

3 Homework Problems

The questions posed (in parenthesis) in this assignment are there to guide your thinking about the problem. You need not record an answer, but do think about your answer to the questions.

Problem 1: The allowed energies for even solutions of the finite square well potential can be determined by finding the roots of the equation:

$$f_{\text{even}}(z) = \sqrt{(z_0/z)^2 - 1} - \tan(z)$$

This is a transcendental equation with no known analytic solution. In this problem, we will solve it numerically.

(A) For $z_0 = \pi/2$ find the root of the f_{even} and compute the corresponding allowed energy. This is the only allowed energy for this narrow and shallow well.

(B) Determine the equation $f_{\text{odd}}(z)$ corresponding to odd solutions, and implement it as python function.

(C) For $z_0 = \pi$, find one root of f_{even} and one root of f_{odd} , and compute the corresponding allowed energies (in units of E_0).

(D) For $z_0 = 3\pi$, find all allowed energies for even and odd solutions.

(E) Use your code to demonstrate that for large z_0 , the first four allowed energies correspond to a kinetic energy:

$$K/E_0 = E + z_0^2$$

which approaches the first four allowed energies of the infinite square well potential. Hint: these corresponding to roots

$$z = \frac{n\pi}{2} \quad n = 1, 2, 3, 4.$$

Problem 2: Use your RK-4 numerical integration to find the even and odd solutions for $z_0 = \pi$. (Are your results consistent with the results of Problem 1?) Plot the normalized wave functions.

Problem 3: In this problem, you will model two finite square wells of width $2a$ with $z_0 = \pi/2$ separated by a distance $2\delta a$ where δ is a dimensionless quantity.

(A) Use your RK-4 numerical integration to find an even solution for $\delta = 3$. Hint: the allowed energy will be near your solution to 1A. Plot the results.

(B) Use your RK-4 numerical integration to find an odd solution for $\delta = 3$. Plot the results. (Do you understand why you now have two solutions, one even and one odd, while in 1A you had only one, even, solution?)

(C) The difference between the energy of the even solution and your answer to Problem 1A can be thought of as a bonding energy. Plot the bonding energy as a function of δ for $\delta = 0$ to $\delta = 3$.

(D) Plot the normalized wave function for two even solutions, one with $\delta = 3$ and one with $\delta = 0.5$.

(E) Plot the normalized wave function for two odd solutions, one with $\delta = 3$ and one with $\delta = 0.5$.